

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI inference performance may slow when multiple users access the system simultaneously.	Increased response time and reduced user experience during peak usage.	High
2	User feedback is currently stored in local JSON files.	Limited scalability and difficult centralized data management.	High
3	Authentication and advanced security mechanisms are not yet implemented.	Limits deployment for enterprise-level applications.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed primarily for demonstration and small-scale usage.	Deploy on cloud platforms with load balancing and auto-scaling to support large numbers of concurrent users.
2	Data Storage	User feedback is stored locally in JSON files.	Migrate to a scalable database such as PostgreSQL or MongoDB for secure and centralized storage.
3	Performance	AI models execute on a single server instance	Optimize model inference, introduce caching, and use GPU-enabled cloud services for faster processing.
4	Security	Basic API communication without user authentication.	Implement secure authentication (OAuth/JWT), HTTPS encryption, role-based access control, and API rate limiting.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment with scalable FastAPI backend and database integration.	1–2 Months	Improved scalability, reliability, and availability.
Phase 2	Personalized user profiles with saved networking history and recommendation engine.	2–3 Months	More customized networking guidance and improved user engagement.
Phase 3	Multi-language support, voice interaction, and mobile application development.	3–5 Months	Increased accessibility and wider user adoption across different regions.
Phase 4	Integration with professional networking platforms and advanced Large Language Models (LLMs) for highly personalized conversation generation.	6+ Months	Enterprise-ready AI networking assistant with enhanced personalization and broader real-world applicability.